

PATENT COOPERATION TREATY

From the
INTERNATIONAL SEARCHING AUTHORITY

PCT

WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY

(PCT Rule 43*bis*.1)

To: PYERNIK PATENT ATTORNEYS P.O. Box 10012 Beer Sheva 8400101 Israel		Date of mailing <i>(day/month/year)</i> 16 Mar 2023	
Applicant's or agent's file reference 361-001PCT		FOR FURTHER ACTION See paragraph 2 below	
International application No. PCT/IL2023/050025	International filing date <i>(day/month/year)</i> 08 Jan 2023	Priority date <i>(day/month/year)</i> 09 Jan 2022	
International Patent Classification (IPC) or both national classification and IPC IPC (20230101) C02F 1/467 B01J 39/07 B01J 45/00 C02F 1/42 C02F 1/461 E04H 4/16 CPC (20130101) C02F 1/4678 B01J 39/07 B01J 45/00 C02F 1/42 C02F 1/46114 E04H 4/16			
Applicant COPPTER WATER TECHNOLOGIES LTD.			

1. This opinion contains indications relating to the following items:

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|-------------------------------------|--------------|--|
| ☒ | Box No. I | Basis of the opinion |
| ☐ | Box No. II | Priority |
| ☐ | Box No. III | Non-establishment of opinion with regard to novelty, inventive step and industrial applicability |
| ☐ | Box No. IV | Lack of unity of invention |
| ☒ | Box No. V | Reasoned statement under Rule 43 <i>bis</i> .1(a)(i) with regard to novelty, inventive step and industrial applicability; citations and explanations supporting such statement |
| ☐ | Box No. VI | Certain documents cited |
| ☐ | Box No. VII | Certain defects in the international application |
| ☒ | Box No. VIII | Certain observations on the international application |

2. FURTHER ACTION

If a demand for international preliminary examination is made, this opinion will be considered to be a written opinion of the International Preliminary Examining Authority ("IPEA") except that this does not apply where the applicant chooses an Authority other than this one to be the IPEA and the chosen IPEA has notified the International Bureau under Rule 66.1*bis*(b) that written opinions of this International Searching Authority will not be so considered.

If this opinion is, as provided above, considered to be a written opinion of the IPEA, the applicant is invited to submit to the IPEA a written reply together, where appropriate, with amendments, before the expiration of 3 months from the date of mailing of Form PCT/ISA/220 or before the expiration of 22 months from the priority date, whichever expires later.

For further options, see Form PCT/ISA/220.

Name and mailing address of the ISA: Israel Patent Office Technology Park, Bldg.5, Malcha, Jerusalem, 9695101, Israel Email address: pctoffice@justice.gov.il	Date of completion of this opinion 09 Mar 2023	Authorized officer JOCHNOWITZ Gershon Telephone No. 972-73-3927146
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**WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY**

International application No.
PCT/IL2023/050025

Box No. I Basis of this opinion

1. With regard to the **language**, this opinion has been established on the basis of:
 - ☒ the international application in the language in which it was filed.
 - ☐ a translation of the international application into _____ which is the language of a translation furnished for the purposes of international search (Rules 12.3(a) and 23.1(b)).
2. ☐ This opinion has been established taking into account the **rectification of an obvious mistake** authorized by or notified to this Authority under Rule 91 (Rule 43*bis*.1(b))
3. With regard to any **nucleotide and/or amino acid sequence** disclosed in the international application, this opinion has been established on the basis of a sequence listing:
 - a. ☐ forming part of the international application as filed.
 - b. ☐ furnished subsequent to the international filing date for the purposes of international search (Rule 13*ter*.1(a)),
☐ accompanied by a statement to the effect that the sequence listing does not go beyond the disclosure in the international application as filed.
4. ☐ With regard to any nucleotide and/or amino acid sequence disclosed in the international application, this opinion has been established to the extent that a meaningful opinion could be formed without a WIPO Standard ST.26 compliant sequence listing.
5. Additional comments:

**WRITTEN OPINION OF THE
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PCT/IL2023/050025

**Box No. V Reasoned statement under Rule 43bis.1(a)(i) with regard to novelty, inventive step and industrial applicability;
citations and explanations supporting such statement**

1. Statement

Novelty (N)	Claims <u>1-27</u>	YES
	Claims _____	NO
Inventive step (IS)	Claims <u>1-27</u>	YES
	Claims _____	NO
Industrial applicability (IA)	Claims <u>1-27</u>	YES
	Claims _____	NO

2. Citations and explanations:

Reference is made to the following documents:

D1: US4525272A

D2: WO2021024938A1

2.1 The subject matter of claims 1-27 isn't described by the above mentioned prior art and therefore is novel and complies with PCT Article 33(2).

2.2 The subject matter of claims 1-27 involves an inventive step and therefore complies with PCT Article 33(3).

A prior art search resulted in publications (see D1-D2) describing water treatment by Cu⁺ ions introduced by electrolytic cells with copper electrodes (similar to those provided by the applicant in the current description's background) or water treatment with ion exchangers aimed to soften water or remove heavy metals, but not a combinations of the two methods that result in a synergistic effect.

While a person of the art could envision alternate Cu⁺ sources, implementing such an alternative is not obvious due to the many inventive steps needed to do so (i.e. choosing the ion exchange method, resin identity, type of chelate, voltage range, temperature, geometry etc.). therefore claims 1-27 are considered inventive (although also see Article 6 deficiency mentioned in box VIII).

2.3 The subject matter of claims 1-27 is considered industrially applicable under Article 33(4) PCT.

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International application No.
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Box No. VIII Certain observations on the international application

The following observations on the clarity of the claims, description, and drawings or on the question whether the claims are fully supported by the description, are made:

The claims define a result to be achieved without detailing how it is achieved as described by the examples in the description, Article 6 PCT. For example, the independent claims don't relate to the physical conditions (temperature, pressure, voltage etc.) of the invention, nor to the identity of all the reactants, their percentage and the relationship between all of the invention's components.